Omar Alchikha

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Education

York University – Lassonde School of Engineering *Toronto, ON*

Bachelor of Engineering, Specialized Honours in Electrical Engineering *Expected May 2028*

- **Upper-Year GPA:** 8.06/9.0; completed courses: Electronic Circuits and Devices (A), Digital Logic Design (A), Electromechanical Energy Conversion (A), Introduction to Power Systems (A+), Electrical Circuits.
- **Fall 2026:** Signals and Systems, Engineering and the Environment; **enrolled for Winter 2027:** Electric Vehicles, Analog Electronics, Linear Control, Power System Analysis, Applied Electromagnetism.

Technical Skills

Circuits & Simulation: circuit analysis, LTspice, PSpice, MATLAB, Python (NumPy/SciPy) modeling, PI loop design, Bode/stability analysis

Hardware & Lab: oscilloscope, DMM, function generator, breadboarding, Arduino Mega, NXP KL43Z, DE10-Lite (Intel MAX 10) FPGA

Programming & Tools: Python, C, Verilog, Java, SQL, Git, Quartus Prime, Microsoft Excel, Microsoft Project

Experience

Project Engineering Intern *Jan 2026 – Present*

De Havilland Aircraft of Canada Limited *Mississauga, ON*

- Reviewed supplier change notices for a **Dash 8-400 weather-radar hardware/software upgrade**, checking affected aircraft configurations and qualification documents and flagging technical discrepancies for engineering follow-up.
- Coordinated engineering deliverables on a **three-aircraft Dash 8-400 refurbishment**, tracking modifications, approvals, and schedule dependencies across avionics, electrical, structures, and interiors.
- Served as Project Engineering focal for customer-specific cabin and safety-equipment modifications, following engineering releases and multidisciplinary sign-offs through to approval.
- Built **Python and SQL** reports on engineering/configuration data to automate deliverable tracking for project schedule and capacity reviews.

Hardware & Systems Support Intern *May 2024 – Sep 2024*

A.B. Academy *Oakville, ON*

- Configured and functionally tested **50+ audio/video and IT devices** across **20+ classrooms** during a school expansion, and wrote setup and troubleshooting guides for staff.

Selected Engineering Projects

Closed-Loop Buck Converter (12 V to 5 V, 2 A) *Simulation, 2026*

- Designed a **100 kHz** buck converter and TL494 analog PI compensator; averaged and switching models in **Python** agree on startup, ripple (33 mV vs. 50 mV spec), and load steps.
- Placed crossover at **7.77 kHz with ∼49° phase margin** at nominal ESR and confirmed stability across 20–100 mΩ ESR; prepared schematic, BOM, and bring-up plan (hardware build and LTspice check pending).

Arc-Fault & Ground-Fault Detection for DC Wiring *Simulation, 2026*

- Designed an analog band-pass/envelope front end and a three-feature detection algorithm in **Python**; Monte-Carlo simulation shows **99.3%** arc detection and 0.33% nuisance trips.
- Ported the detector to **C99** Arduino firmware matching the Python reference on 36/36 test traces; sensing board designed, not yet built.

FPGA Digital Control Core – Verilog, Intel MAX 10 *RTL + simulation, 2026*

- Designed a **Verilog** PWM generator (20 kHz, 10-bit, glitch-free duty updates) and 4× quadrature decoder with input synchronizers, glitch filter, and illegal-transition flag.
- Verified both with self-checking **Icarus Verilog** testbenches against independent models (**23/23 checks pass**); prepared a Quartus Prime project with pin map and timing constraints (not yet on the board).

DC Motor Speed Control – System Identification & PI *Simulation, 2026*

- Built a Python pipeline that fits a motor model to step-response data and designs a PI speed loop (60° phase margin, ≥55.9° with ±30% plant variation); validated end to end on **synthetic** data (90.7% fit).
- Wrote Arduino firmware for the 100 Hz PI loop with 4× quadrature-encoder decoding and anti-windup (cut simulated overshoot from 31% to 5.6%); not yet run on a motor.